Title: Anomaly Detection Pipelines: Toward High-dimensional Feature Selection in Urban Mobility Analysis

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Abstract:

High-dimensional feature selection remains a critical challenge for data warehousing, particularly in urban mobility analysis. This paper proposes a framework built on anomaly detection pipelines that explicitly accounts for reproducibility of analytical workflows. We evaluate the framework on open census microdata and compare it against 4 baseline methods. Experimental results show that the proposed approach improves forecast MAPE by 22.1% while keeping AUC-ROC within acceptable bounds. We further analyse its limitations and discuss implications for practical deployment.

1. Introduction

The rapid maturation of data warehousing has changed how researchers and practitioners approach high-dimensional feature selection, yet practical deployment is still limited by scalability, cost, and reproducibility concerns. Over the past decade, anomaly detection pipelines have attracted sustained attention as a promising route toward solving high-dimensional feature selection in realistic settings such as urban mobility analysis. Despite considerable progress, existing solutions for high-dimensional feature selection tend to degrade when moving from controlled laboratory conditions to production workloads, which motivates a more systematic investigation. A central difficulty is that improvements in silhouette coefficient frequently come at the expense of RMSE, forcing designers to accept trade-offs that are poorly understood.

This paper addresses these gaps by studying how anomaly detection pipelines interact with probabilistic topic models when applied to high-dimensional feature selection. Our work is motivated by the growing demand for urban mobility analysis, where reliability and transparency requirements are especially strict. Understanding reproducibility of analytical workflows is equally important, because it shapes both the theoretical limits and the operational cost of any proposed approach. The remainder of this paper reviews related work, describes the proposed methodology, and concludes with a discussion of limitations and directions for future research.

2. Related Work

Dubois et al. (2016) introduced an early framework based on anomaly detection pipelines, reporting a 37.3% improvement in AUC-ROC over previous baselines. Subsequent work by Hoffmann et al. (2018) extended this line of research to urban mobility analysis, although the evaluation was restricted to small-scale settings. A survey by Hoffmann et al. (2012) categorised existing approaches to high-dimensional feature selection into three families and highlighted the lack of standard benchmarks. Researchers working on data warehousing have explored probabilistic topic models as an alternative, with Chen et al. (2012) noting that the approach scales poorly beyond 32 components. More recently, Nakamura et al. (2017) combined anomaly detection pipelines with probabilistic topic models, achieving a 21.1% reduction in overhead on open census microdata.

Several studies have criticised the reliance on synthetic workloads, arguing that results on open census microdata may not generalise to urban mobility analysis. In contrast to these efforts, our work treats high-dimensional feature selection and reproducibility of analytical workflows jointly rather than as independent concerns. Industry reports corroborate these academic findings, pointing to high-dimensional feature selection as a persistent obstacle for urban mobility analysis. Despite this body of literature, no prior study has systematically compared anomaly detection pipelines and probabilistic topic models under identical

experimental conditions.

3. Methodology

We formalise high-dimensional feature selection as a constrained optimisation task in which the objective is to maximise forecast MAPE while keeping AUC-ROC within a prescribed budget. The proposed architecture consists of four stages: data acquisition, preprocessing, core processing based on anomaly detection pipelines, and a validation layer that incorporates probabilistic topic models. All experiments were conducted on open census microdata, which provides 128 labelled instances or configurations covering a broad range of operating conditions. To ensure reproducibility, every experiment was repeated 30 times with independent random seeds, and we report mean values together with 95% confidence intervals.

Hyperparameters were selected through grid search over a validation split, and the final configuration was fixed before any evaluation on held-out data. The core routine applies anomaly detection pipelines to reduce the effect of reproducibility of analytical workflows, followed by a calibration step that stabilises AUC-ROC. Our implementation is written in Python and C++, and was executed on a workstation with 64 GB of memory and a single mid-range accelerator. We compare against 4 baseline methods, including a conventional approach built on probabilistic topic models and two recent variants reported in the literature.

Ablation experiments show that removing the probabilistic topic models component reduces forecast MAPE by approximately 13.6%, confirming its contribution. A sensitivity analysis over 500 parameter settings shows that performance remains stable, with variation in silhouette coefficient below 14.6%. Threats to validity include the limited diversity of open census microdata and the possibility that deployment in urban mobility analysis introduces conditions not captured in our setup.

4. Conclusion

This paper investigated high-dimensional feature selection through the lens of anomaly detection pipelines, and demonstrated that a carefully designed pipeline can improve forecast MAPE by 22.1% on open census microdata. The findings suggest that data warehousing will continue to benefit from closer integration of anomaly detection pipelines and probabilistic topic models, particularly in urban mobility analysis. Several limitations remain, including the restricted scale of our evaluation and the simplified treatment of reproducibility of analytical workflows. Future work will extend the framework to larger deployments, explore tighter theoretical guarantees, and release an open benchmark to facilitate fair comparison. We hope the evidence presented here encourages the community to adopt more rigorous and transparent evaluation practices for high-dimensional feature selection.